

REMARKS ON SIX-TANGENTIAL CONICS TO SMOOTH PLANE SEXTICS

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ABSTRACT.

Let k be a positive integer. We consider the complete graph with vertex set $V_k = \{1, \dots, k\}$ and edge set

$$E_k := \{[i, j] \mid i, j \in V_k, i < j\}.$$

Let m be a positive integer, and we put

$$M := \{-m, -m+1, \dots, m\}.$$

Let $\mathcal{W}_k$ be the set of weights $w: E_k \rightarrow M$, that is, $\mathcal{W}_k$ is the set of weighted complete graphs on V_k . Then the symmetric group $\mathfrak{S}_k$ act on V_k from the right. Then $\mathcal{S}_k$ acts on $\mathcal{W}_k$ from the left. We also define Sw_k to be the multiplicative group of all mappings $s: V_k \rightarrow \{\pm 1\}$, which acts on $\mathcal{W}_k$ by switching, that is,

$$(sw)([i, j]) = s(i)s(j)w([i, j])$$

for $s \in \text{Sw}_k$. Then the semidirect product

$$G_k := \text{Sw}_k \rtimes \mathfrak{S}_k$$

acts on $\mathcal{W}_k$ from the left. The purpose of this section is to describe an efficient algorithm for obtaining the list of orbits of the action of G_k on $\mathcal{W}_k$.

First, we introduce a total order on the set E_k of edges by

$$[i, j] < [i', j'] \iff j < j' \text{ or } (j = j' \text{ and } i < i').$$

Then we express elements w of $\mathcal{W}_k$ by arranging the values $w([i, j])$ in this order:

$$w = (w([1, 2]), w([1, 3]), w([2, 3]), w([1, 4]), \dots).$$

Let $|w|$ denote the vector obtained from w by taking the absolute value of each component. We put $|M| := \{0, 1, \dots, m\}$. Then we have a mapping

$$(0.1) \quad \mathcal{W}_k \longrightarrow |\mathcal{W}_k| := |M|^{E_k}, \quad w \longmapsto |w|.$$

Notice that $\mathfrak{S}_k$ acts on $|\mathcal{W}_k|$, and the mapping (0.1) is $\mathfrak{S}_k$ -equivariant. Moreover, the action of Sw_k on $\mathcal{W}_k$ preserves each fiber of (0.1). We then define a total order on $\mathcal{W}_k$ by

$$w < w' \iff |w| <_{\text{lex}} |w'| \text{ or } (|w| = |w'| \text{ and } w <_{\text{lex}} w').$$

We say that $w \in \mathcal{W}_k$ is G_k -minimal if w is minimal in its orbit wG_k . We obtain the list of G_k -orbits in $\mathcal{W}_k$ by enumerating all G_k -minimal elements in $\mathcal{W}_k$. By definition, we have the following:

$$\begin{aligned} w \text{ is } G_k\text{-minimal} \iff & \begin{aligned} & |w| \text{ is minimal with respect to } <_{\text{lex}} \text{ in its } \mathfrak{S}_k\text{-orbit} \\ & \mathfrak{S}_k|w| \text{ in } |\mathcal{W}_k|, \text{ and } w \text{ is minimal with respect to} \\ & <_{\text{lex}} \text{ in its } \text{Sw}_k\text{-orbit } \text{Sw}_k w. \end{aligned} \end{aligned}$$

By this observation, to determine whether an element $w \in \mathcal{W}_k$ is G_k -minimal, it suffices to make at most $k! + 2^k$ comparisons, which is much smaller than the size $2^k k!$ of G_k .

Suppose that $k' < k$. By taking the prefix of length $k'(k'-1)/2$ of each $w \in \mathcal{W}_k$, we obtain a mapping

$$\mathcal{W}_k \longrightarrow \mathcal{W}_{k'}, \quad w \longmapsto w|_{k'}.$$

Lemma 0.1. *Suppose that $w \in \mathcal{W}_k$ is G_k -minimal. Then $w|_{k'}$ is $G_{k'}$ -minimal.*

Proof. We have a natural embedding $G_{k'} \hookrightarrow G_k$ denoted by $g' \mapsto \tilde{g}'$ by

$$i^{\tilde{\tau}'} := \begin{cases} i^{\tau'} & \text{if } i \leq k', \\ i & \text{if } i > k', \end{cases} \quad \tilde{s}'(i) := \begin{cases} s'(i) & \text{if } i \leq k', \\ i & \text{if } i > k', \end{cases}$$

for $\tau' \in \mathfrak{S}_{k'}$ and $s' \in \text{Sw}_{k'}$. □

Using this lemma, we construct the list of G_k -minimal elements by induction on k .

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